

2 — Derived Values: \$derived

COMPUTED VALUES

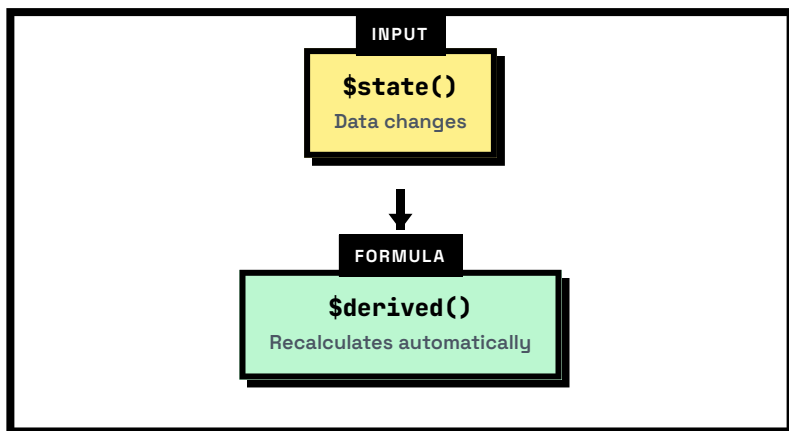
Q. How do you compute a value based on other reactive state so that it updates automatically without manual recalculation?

ANSWER. When building a shopping cart, you need the **totalPrice** to update automatically whenever items are added or removed. Instead of manually recalculating the total every time, you use the **\$derived** rune to create an auto-updating formula. Svelte caches the result and only re-runs the math when the underlying items actually change.

App.svelte

```
<script>
  let count = $state(1);
  let doubled = $derived(count * 2);

  // Derived from objects
  let user = $state({ first: 'Ada', last: 'Lovelace'
});
  let full = $derived(`${user.first} ${user.last}`);
</script>
```



i Summary. Think of **\$derived** as an auto-updating formula. This is powerful because it calculates a value once and caches it. It only re-runs when the inputs actually change. This keeps your app

blazing fast without manual recalculation logic.